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Gaming machine and control method for operating gaming machines to display computer animated reel games

Abstract

A gaming machine including a cabinet, a display device mounted to the cabinet, and a control unit operably coupled to the display device is described herein. The control unit includes a processor programmed to execute an algorithm to render an animated sequence of computer-generated images on the display device including the steps of rendering a primary game screen on the display device including a plurality of virtual reels displayed within a reel window including a plurality of cells arranged in a plurality of rows and columns. The processor transforms the reel window by animating the reel window to increase a size of the reel window to display an additional row of cells within the reel window upon determining the game outcome displays a special symbol within the reel window. The processor also decreases the size of the reel window upon determining the game outcome does not display the special symbol.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application claims the benefit of U.S. Provisional Patent Application Ser. No. 63/358,780, filed Jul. 6, 2022, the disclosure of which is hereby incorporated by reference in its entirety for all purposes.

TECHNICAL FIELD

(1) The present invention relates to a gaming machine, a control method for a gaming machine, and a program for a gaming machine.

BACKGROUND

(2) A gaming machine represented by a slot machine is highly popular among casino customers as a device that provides gaming that is easy to enjoy, and recent statistics report that sales from gaming machines account for the majority of casino earnings. Initial slot machines were simple devices, wherein an inserted coin is received, a configured reel rotates and stops mechanically according to a handle operation, and a win or a loss is determined by a combination of symbols stopped on a single pay line. However, recent gaming machines, such as mechanical slot machines driven by a highly accurate physical reel via a computer controlled stepping motor, video slot machines that display a virtual reel on a display connected to a computer, and various gaming machines that apply similar technology to other casino games are quickly advancing. For the manufacturers that develop these gaming machines, an important theme is to provide an attractive game that strongly attracts casino customers as players, and improves the functionality of the gaming machine.

SUMMARY OF THE INVENTION

(3) In one aspect of the present invention, a gaming machine is provided. The gaming machine

includes a cabinet, a display device mounted to the cabinet, and a control unit operably coupled to the display device. The control unit includes a processor programmed to execute an algorithm to render an animated sequence of computer-generated images on the display device including the steps of rendering a primary game screen on the display device including a plurality of virtual reels displayed within a reel window including a plurality of cells arranged in a plurality of rows and columns. The processor initiates an instance of a computer-animated primary reel game by animating the plurality of virtual reels to simulate spinning the plurality of virtual reels through the reel window and sequentially stopping the plurality of virtual reels to display a game outcome. The processor transforms the reel window by animating the reel window to increase a size of the reel window to display an additional row of cells within the reel window for use in a subsequent instance of the computer-animated primary reel game, upon determining the game outcome displays a special symbol within the reel window. The processor also transforms the reel window by animating the reel window to decrease the size of the reel window to display a fewer number of rows of cells within the reel window for use in the subsequent instance of the computer-animated primary reel game, upon determining the game outcome does not display the special symbol.

(4) In another aspect of the present invention, a method of operating a gaming machine is provided. The gaming machine includes a cabinet, a display device mounted to the cabinet, and a control unit including a processor operably coupled to the display device. The method includes the processor performing an algorithm to render an animated sequence of computer-generated images on the display device including the steps of rendering a primary game screen on the display device including a plurality of virtual reels displayed within a reel window including a plurality of cells arranged in a plurality of rows and columns. The processor initiates an instance of a computer-animated primary reel game by animating the plurality of virtual reels to simulate spinning the plurality of virtual reels through the reel window and sequentially stopping the plurality of virtual reels to display a game outcome. The processor transforms the reel window by animating the reel window to increase a size of the reel window to display an additional row of cells within the reel window for use in a subsequent instance of the computer-animated primary reel game, upon determining the game outcome displays a special symbol within the reel window. The processor also transforms the reel window by animating the reel window to decrease the size of the reel window to display a fewer number of rows of cells within the reel window for use in the subsequent instance of the computer-animated primary reel game, upon determining the game outcome does not display the special symbol.

(5) In yet another aspect of the present invention, a non-transitory computer-readable storage media having computer-executable instructions embodied thereon to operate a gaming machine is provided. The gaming machine includes a cabinet, a display device mounted to the cabinet, and a control unit including a processor operably coupled to the display device. When executed by the processor, the computer-executable instructions cause the processor to perform an algorithm to render an animated sequence of computer-generated images on the display device including the steps of rendering a primary game screen on the display device including a plurality of virtual reels displayed within a reel window including a plurality of cells arranged in a plurality of rows and columns. The processor initiates an instance of a computer-animated primary reel game by animating the plurality of virtual reels to simulate spinning the plurality of virtual reels through the reel window and sequentially stopping the plurality of virtual reels to display a game outcome. The processor transforms the reel window by animating the reel window to increase a size of the reel window to display an additional row of cells within the reel window for use in a subsequent instance of the computer-animated primary reel game, upon determining the game outcome displays a special symbol within the reel window. The processor also transforms the reel window by animating the reel window to decrease the size of the reel window to display a fewer number of rows of cells within the reel window for use in the subsequent instance of the computer-animated primary reel game, upon determining the game outcome does not display the special symbol.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1A is a perspective view of a gaming machine, according to an embodiment of the present invention.
- (2) FIGS. 1B-1G are perspective views of the gaming machine, according to other embodiments of the present invention.
- (3) FIG. 2 is a functional block diagram of the gaming machine in FIGS. 1A-1G.
- (4) FIG. 3 is an illustration of a game displayed on a display area of the gaming machine, according to an embodiment of the present invention.
- (5) FIG. 4 is an illustration of virtual reels that may be used with the game shown in FIG. 3, according to an embodiment of the present invention.
- (6) FIGS. 5-8 are block diagrams of a game control unit that may be used with to perform the function of executing a game on the gaming machine.
- (7) FIG. 9 is a functional block diagram of a server computer system, according to an embodiment of the present invention.
- (8) FIG. 10 is a front view of a mobile computing device that may be used with the server computer system of FIG. 9.
- (9) FIG. 11 is an illustration of exemplary reel strip data file for use in generating virtual reels shown in FIG. 4 and FIGS. 17-23 with symbol arrangements showing the order of symbols displayed on the display area of the game, according to an embodiment of the present invention.
- (10) FIG. 12 is an illustration of a reel stop position data file that may be used to execute the game, according to an embodiment of the present invention.
- (11) FIG. 13 is an illustration showing one example of a pay line set on the determination area of the game.
- (12) FIGS. 14-16 are flow charts illustrating algorithms used during operation of the gaming machine and/or the mobile computing device to generate and render animated sequences of computer-generated images on graphic user interface screens to display the game to the player, according to one embodiment of the present invention.
- (13) FIGS. 17-27 are exemplary illustrations of animated sequences of computer-generated images of game screens illustrating the algorithms shown in FIGS. 14-16.
- (14) Corresponding reference characters indicate corresponding parts throughout the drawings.

DETAILED DESCRIPTION OF EMBODIMENTS

(15) With reference to the drawings and in operation, the present invention is directed towards a gaming machine, a control method for a gaming machine, and a program for a gaming machine and/or mobile computing device that provides a game to a player. The gaming machine according to the present embodiment, receives a predetermined game value from the player, generates a game result, and provides a payout to the player according to the game result and one or more pay tables. The gaming machine includes a gaming cabinet that includes a display device and a user input device. The display device includes a display configured to display computer generated graphics. The user input device includes a plurality of input buttons that allows a player to initiate game play and interact with video images being displayed on the display device. The gaming machine includes a control unit having a computer processor that is programmed generate and display a structured graphic user interface displayed on the display device using computer generated graphics. The structure graphic user interface includes a plurality of virtual reels displaying game symbols within a grid. The control unit is also programmed to animate the virtual reels to simulate spinning the virtual through the grid, and stop the virtual reels to reveal an outcome of a game. The control unit is also programmed to generate and render animated sequences of computer-generated images on graphic user interface screens to display bonus features associated with the game.

(16) Referring to FIGS. 1A-1D, in the illustrated embodiment, the gaming machine **10** includes a cabinet **12** providing a display unit **14** and a control panel **16** and may also house a player tracking or ranking unit **18**. The cabinet **12** also houses a game control unit **20** (see FIG. 2) that controls each part (see below). The control unit **20** also implements a random number generator (RNG) that is used during operation of the game. Each configuration is described below.

(17) The display unit **14** may be liquid crystal display devices and/or organic EL display devices and the like, and by controlling via each control unit **20**, the display area mentioned below functions as a display unit **14** provided to the player. In some embodiments, the display unit **14** may include an upper display **22** and a lower display **24**. As shown FIGS. 1A, 1F, and 1G, in some embodiments, the upper and lower displays **22**, **24** are shown on a single display device. In other embodiments, as shown in FIGS. 1B-1E, the upper and lower displays **22**, **24** may be shown on multiple display devices.

(18) Speakers **26** are provided on the left and right of the cabinet **12**, and by controlling via the control unit **20**, sound is provided to the player. On the control panel **16**, a bill/ticket identification device **28**, a printer device **30**, and an operation unit **32** are provided.

(19) The player tracking unit **18** may be housed on the center of the front surface of the cabinet **12** below the display unit **14**. The player tracking unit **18** has a card reader that recognizes a player identification card, a display that presents data to the player, and a keypad that receives input by the player. This type of player tracking unit **18** reads information recorded on the player identification card inserted by the player into the card reader, and displays the information and/or information acquired by communicating with the external system on the display, by cooperatively operating with the control unit **20** mentioned below or an external system. Further, input from the player is received by the keypad, the display is changed according to the input, and communication with the external system is carried out as necessary.

(20) The bill/ticket identification device **28** is disposed on the control panel **16** in a state where the insertion opening that a bill/ticket is inserted into is exposed, an identification part that identifies a bill/ticket by various sensors on the inside of the insertion opening is provided, and a bill/ticket storage part is provided on the outgoing side of the identification part. The bill/ticket identification device **28**, receives and identifies bills/tickets (including vouchers and coupons) that are the game value as a game executing value, and notifies the control unit **20** mentioned below.

(21) The printer device **30** is disposed on the control panel **16** in a state where the ticket output opening that a ticket is output from is exposed, a printing part that prints predetermined information on a printing paper on the inside of the ticket output opening is provided, and a housing part that houses the printing paper inside the paper inlet side of the printing part is provided. The printer device **30**, under the control of the control unit **20** mentioned below, prints information on paper and outputs a ticket according to credit payout processing from the gaming machine **10**. The output ticket can use the payout credit as game play by being inserted into the bill/ticket identification device **28** of another gaming machine, or, can be exchanged for cash by a kiosk terminal inside of the casino or a casino cage.

(22) The operation unit **32** receives the operation of the player. The operation unit **32** includes a group of buttons **34** that receives various instructions from the player on the gaming machine **10**. The operation unit **32**, for example, may include a spin button and a group of setting buttons. The spin button receives an instruction to start (start rotating the reel) an instance of the game. The group of setting buttons **34** includes a group of bet buttons, a denomination button, a group of line-designation buttons, a max bet button, and a payout button and the like. The group of bet buttons receives an instruction operation regarding the bet amount of credits (bet number) from the player. The group of line-designation buttons receive an instruction operation that designate a pay line subjected to a line judgment below from the player. The max bet button receives an instruction operation regarding the bet of the maximum amount of credits that can be bet at one time from the player. The payout button receives an instruction operation instructing a credit payout accumulated

in the gaming machine **10**. The gaming machine **10** also includes illumination devices **36** that provides decorative lighting to the gaming machine **10**.

(23) In one embodiment, the control panel **16** includes a plurality of user input devices that may include an acceptor device which accepts media associated with a monetary value to establish a credit balance, a validator configured to identify the physical media, a cash-out button actuable to cause an initiation of a payout associated with the credit balance. The acceptor device may include a touchscreen display associated with the display unit **14** and/or the player tracking unit **18**, the bill/ticket identification device **28**, the operation unit **32**, the player tracking unit **18**, a coin slot, a ticket in ticket out (TITO) system, a bill acceptor, and/or any suitable device that enables the gaming machine **10** to receive media associated with a monetary value and establish a credit balance for use in playing the gaming machine **10**. In one embodiment, the acceptor device may be configured to receive physical media such as, for example, a coin, a medal, a ticket, a card, a bill, currency, and/or any suitable physical media that enables the gaming machine **10** to function as described herein. The acceptor device may also be configured to accept virtual media such as, for example, a player tracking account, a virtual credit balance, reward points, gaming credits, bonus points, and/or any suitable virtual media that enables the gaming machine **10** to function as described herein.

(24) For example, in one embodiment, the coin slot may include an opening that is configured to receive coins and/or tokens deposited by the player into the gaming machine **10**. The control unit **20** converts a value of the coins and/or tokens to a corresponding amount of gaming credits that are used by the player to wager on games played on the gaming machine **10**. The bill acceptor may include an input and output device that is configured to accept a bill, a ticket, and/or a cash card into the bill acceptor to enable an amount of gaming credits associated with a monetary value of the bills, ticket, and/or cash card to be credited to the gaming machine **10**. In one embodiment, the bill acceptor also includes a printer (not shown) that is configured to dispense a printed voucher ticket that includes information indicative of an amount of credits and/or money paid out to the player by the gaming machine **10** during a gaming session. The voucher ticket may be used at other gaming devices, or redeemed for cash, and/or other items as part of a casino cashless system.

(25) FIG. 2 shows a functional block diagram of the gaming machine **10**, according to the present embodiment. The gaming machine **10** provides the control unit **20**. Within the cabinet **12**, control unit **20** includes a control board equipped with a central processing unit **38** (abbreviated as CPU below) including a processor, an interface unit (or part) **40**, a memory device including a memory **42** and a storage **44**, and the like are incorporated. The control board is configured so that communication is possible through the interface unit **40** and each of the components equipped on the cabinet **12**. The CPU **38** controls the operation of each component by executing the program recorded in the memory **42** or the storage **44** of the CPU **38**, and provides a game to the player. The function of the CPU **38** is to execute and display the game on the display unit **14** of the gaming machine **10**.

(26) The control unit **20** is configured as the interface unit **40** including a chip set providing communication functions of the CPU **38**, a memory bus connected to a CPU **38**, various expanding buses, serial interfaces, USB interfaces, Ethernet (registered trademark) interfaces and the like, and a computer unit where the CPU **38** provides the addressable memory **42** and the storage **44** through the interface unit **40**. The memory **42** can be configured to include RAM that is a volatile storage medium, ROM that is a nonvolatile storage medium, and EEPROM that is a rewritable nonvolatile storage medium. The storage **44** provides the control unit **20** as an external storage device function, can use reading devices such as a memory card that is a removable storage medium, and a magneto optical disk and the like, and can use hard disks.

(27) On the interface unit **40**, in addition to the CPU **38**, the memory **42**, and the storage **44**, a bill/ticket identification unit controller **46**, a printer unit controller **48**, the player tracking unit **18**, a graphic controller **50**, an input controller **52**, and a sound controller **54** are connected. That is, the

control unit **20** is connected to the operation unit **32** through the input controller **52**, and connected to the display unit **14** through the graphic controller **50**. Further, when illumination devices **36** that provides decorative lighting to the gaming machine **10** is provided, the illumination is controlled under the control of the control unit **20** on the interface unit **40**, and an illumination controller **56** that controls the illumination devices **36** to provide a decorative lighting effect may be connected.

(28) The control unit **20**, which includes memory **42** and storage **44**, controls each part by executing a program stored in the memory **42** and the storage **44**, and provides a game to the player. Here, for example, the memory **42** and storage **44** may be configured to store a program and data of an operating system and subsystem that provide the basic functions of the control unit **20** to the EEPROM of the memory **42**, and stores a program and data of an application that provides a game to the storage **44**. According to such a configuration, it can be easy to change or update a game by replacing the storage **44**. Further, the control unit **20** may be a multiprocessor configuration that has a plurality of CPUs.

(29) Each block connected to the control unit **20** is described below. The bill/ticket identification unit controller **46** operates the bill/ticket identification device **28** to receive bills/tickets in the insertion opening, and notifies the control unit **20** of identifying information corresponding to the assortment of bills or the payout processing of credits. The bill/ticket identification unit controller **46** notifies the information to the control unit **20**, and the control unit **20** increases the usable credit amount inside of the game according to the notified content. The printer unit controller **48** corresponds to the printer device **30**, and under the control of the control unit **20** that receives an operation of the payout button of the group of setting buttons **34**, information corresponding to the credit payout processing from the gaming machine **10** is printed and output on a printed ticket.

(30) The player ranking (or tracking unit) unit **18** cooperatively operates with the control unit **20**, and sends and receives information and the like of the player from the casino management system. The graphic controller **50** controls the display unit **14**, under the control of the control unit **20**, and displays a display image that includes various graphic data. The sound controller **54** drives the speakers **26** under the control of the control unit **20**, and provides various sounds such as an announcement, sound effects, BGM and the like.

(31) Further, the interface unit **40**, has various communication interfaces for communicating with the exterior of the gaming machine **10**, for example the interface unit **40** can communicate with an external network by Ethernet **58**, **60**, and a serial interface **62**. In the present embodiment, one example shows when there is communication between a well-known server-side gaming network **58** (Server Based Gaming), a G2S network **60** (Game to System), and a slot information system **62** (Slot Data System), respectively.

(32) FIG. 3 schematically shows a display area **64** provided by the gaming machine **10**. Such a display area **64** is displayed on the display unit **14** (the upper display **22** and/or the lower display **24**) by the control unit **20** executing a predetermined program. In the illustrated embodiment, the display area **64** is displayed on the lower display **24**. For instance, as shown, during a game, the upper display **22** may be utilized to display game related information, e.g., game title information and/or graphics.

(33) In one aspect of the present invention, the control unit **20** executes a game execution program to provide a game **66** that includes a primary game **68** and one or more bonus feature events **69** (shown in FIGS. 22-23). The control unit **20** displays a game screen within the display area **64** that includes the primary game **68** and the bonus feature events. In the illustrated embodiment, the primary game **68** and the bonus feature events are displayed on the lower display **24**. The upper display **22** may be used to display animations and/or game identifying information during the game and/or during an attract mode. Further, the lower display **24** may display a decorative area, and an area that displays credit amount, bet number, and a credit amount obtained by winning (WIN number) and the like. In another embodiment, the primary game **68** may be displayed in the lower display **24** and the bonus feature events may be displayed in the upper display **22**.

(34) In the illustrated embodiment, the primary game **68** includes a video slot game using a plurality of virtual reels **70, 72, 74, 76, 78** (shown in FIGS. **3-4**). The video slot game utilizes a grid **80** in the display area **64**. The illustrated embodiment shows the state of displaying the display area **64** in the lower display **24**. By using such a display area, the gaming machine **10** of the present embodiment operates as a slot machine that pays a payout according to a winning combination of symbols displayed on the display area **64**.

(35) The display unit **14** displays a plurality of game symbols in the grid **80**. The grid **80** has a plurality of rows (r) and columns (c). The grid **80** is configured by a plurality of cells **82** that are the stop position of symbols. On each of the plurality of cells **82** of the display area **64**, one symbol is stopped and displayed.

(36) In the illustrated embodiment, the grid **80** is displayed with fifteen cells **82** arranged in three rows and five columns, and the control unit **20** generates five virtual reels **70, 72, 74, 76, 78** for use in displaying the primary game **68**. On each cell **82** of the grid **80**, as shown in FIG. **3**, a game symbol **84** is displayed based on the symbol arrangement of virtual reels including virtual reels **70, 72, 74, 76, 78** (also shown in FIG. **4**) configured as a virtual reel set **86**. That is, the cells **82** of the grid **80** correspond to the virtual reels **70** to **78**, by column, and the game symbols **84** disposed on predetermined parts of each virtual reels **70** to **78** are displayed. In the illustrated embodiment, each of the virtual reels **70** to **78** displays multiple symbols within a corresponding column with the virtual reel in a stopped position. For example, as shown in FIG. **3**, when stopped, each virtual reel **70** to **78** displays game symbols within each corresponding column. Furthermore, by moving (scrolling or spinning) each game symbol **84** by column based on the symbol arrangement of the virtual reels **70** to **78**, the game symbols **84** displayed in the cells **82** of the grid change, and by stopping the movement (scrolling or spinning) by columns, the game symbols **84** are stopped. Here, the virtual reels **70** to **78** are data where the control unit **20** uses a program having the memory **42** or the storage **44**, and data showing the symbol arrangement (i.e., the order of symbols on each reel strip) regulated by each cell column. Further, the virtual reel set **86** is a general term for such virtual reels **70** to **78**.

(37) In some embodiments, the control unit **20** generates each virtual reel **70** to **78** using reel strip data files **88** (shown in FIG. **11**) and symbol image data files stored in memory **42** and/or storage **44**. The reel strip data file **88** includes a reel designation **90** associated with each virtual reel **70** to **78** and sequential symbol position logic cells **92** associated with each reel designation. Each symbol position logic cell **92** includes indicators for rendering the virtual reels with a plurality of game symbols. The reel strip data file **88** includes information associated with a plurality of reel strips **94, 96, 98, 100, 102** that correspond to each virtual reel **70** to **78**, and are used to generate virtual reel **70** to **78**. Each reel strip **94, 96, 98, 100, 102** includes a number of symbol positions **104** configured to be populated by game symbols **84** that are selected from symbol image data files stored in memory **42** and/or storage **44**. The reel strips **94-102** also include reel stop positions **106** associated with each symbol position **104**. The symbol positions **104** and the corresponding game symbols **84** are arranged in an order defined by each reel strip **94, 96, 98, 100, 102**. In addition, one or more reel strips **94, 96, 98, 100, 102** includes a different number of symbol positions **104**.

(38) In some embodiments, each reel strip **94, 96, 98, 100, 102**, includes a plurality of fixed symbol positions **104** for displaying game symbols **84**. Each fixed symbol position **104** includes instructions for displaying a predefined game symbol **84** selected from a set of game symbols **84** included in a game symbol image data file. For example, for each play of the game, the fixed symbol positions **104** have an associated predefined game symbol **84** from a symbol set. The game symbol image data file includes the details of game symbols **84** populating the fixed symbol positions **104** and includes varieties of game symbols **84**. This symbol set includes card symbols ("9", "10", "J", "Q", "K", and "A") that imitate playing cards as regular symbols, and picture symbols ("PicA", "PicB", "PicC", and "PicD") that show a pattern. Further, this symbol set includes a wild symbol ("Wild") that is substituted as another symbol when a win combination is

determined and a special symbol (“Special”) that may be used to determine if a game feature is to be provided. Each of these symbols have a different rank from each other regarding their value when winning, their rank gradually raises in this order: “9”, “10”, “J”, “Q”, “K”, “A”, “PicE”, “PicD”, “PicC”, “PicB”, “PicA”. A combination of symbols that includes high-ranking symbols when winning, can obtain a larger winning payout compared to a combination of low-ranking symbols when winning.

(39) One or more of the reel strips **94, 96, 98, 100, 102** may also include a plurality of variable symbol positions **104** that display a varying inner symbol (“inner”). The variable symbol positions **104** are populated using a variable symbol reel strip data file and/or a variable symbol weight table. The variable symbol reel strip data file includes a variable symbol reel strip that includes a number of symbol positions populated by game symbols that are selected from game symbol image files, and stop positions associated with each symbol position. When generating the virtual reels **70 to 78**, the control unit **20** is programmed to populate the variable symbol positions **104** by randomly selecting game symbols from the predefined group of symbols using the variable symbol reel strip. For example, the control unit **20** may be programmed to randomly select a stop position of the variable symbol reel strip, select the symbol position associated with the randomly selected stop position, and populate the variable symbol position **104** with the game symbol **84** associated with the selected symbol position.

(40) It should be noted that in one aspect of the present invention, one or more dynamic virtual reel strips may be utilized. Using virtual reel strips, the symbols and/or symbol positions and/or virtual reel strips and/or length or size and/or any aspect of a virtual reel strip may change from one spin or play to the next. For example, a dynamic reel strip includes a plurality of symbol positions with symbols from the symbol set and a plurality of variable symbol positions. The variable symbol positions may be in the form of one or more stacks, i.e., adjacent symbol positions. In one embodiment, the location and/or size of the stacks may change from one spin to the next, either randomly and/or in a predetermined pattern.

(41) Alternatively, a virtual reel strip associated with a column of cells may be dynamically changed from one spin or play to another spin or play. This, may occur randomly, every spin or play and/or in a predetermined pattern. It should be noted that in the illustrated embodiment, each column of the grid **80** has a corresponding reel. When the reel stops, a symbol from the respective reel appears in each one of the cells of the respective column of the grid **80**.

(42) In general, the control unit **20** starts an instance of the primary game **68** and determines the stop position of each virtual reels **70 to 78** randomly using a reel stop position data file **108** (shown in FIG. 12) stored in the memory device. The virtual reels **70 to 78** that are displayed in the display unit **14** (for example, the lower display **24**) are moved from a current position, and stopped based on a stop position to express an outcome of the primary game. Due to this, in the display or grid **80**, the symbols included on the virtual reels **70 to 78** are continuously moved (scrolled or spun) in a vertical direction of the display area **64**, and one symbol of one cell **82** is aligned in an order of the symbol based on the symbol arrangement is stopped so that it is displayed.

(43) The control unit **20** changes and stops the plurality of symbols displayed on the display unit **14** according to the operation of the player received by the operation unit **32**, and a payout may be paid according to the stopped symbols inside the display area **64** and a pay line set and payable data file **110** (shown in FIG. 13) stored in the memory device.

(44) In the display area **64**, a pay line is set that is used when winning is determined. The pay line is set to be extended over the column on the right end from the cells of the column of the left end, and is a line that combines the plurality of cells **82** determining a win. The number of effective lines within the set pay line is selected by the operation of a group of line designation buttons included in the group of setting buttons **34** of the operation unit **32** for the player. The control unit **20**, in regard to the result of a game that is a combination of symbols, determines a win when a predetermined number of identical symbols is surpassed and aligned on a set pay line, and pays a

payout to the player according to the type and number of symbols. The system for determining a win may determine a win when a predetermined number of identical symbols from cells of the column on the left end are aligned on a set pay line, may determine a win when a predetermined number of identical symbols from cells of the column on the right end are aligned on a set pay line, and may determine a win when a predetermined number of identical symbols are aligned on a continuous column on a predetermined pay line. It should be noted that pay lines shown other than (or in addition to) the pay lines shown in FIG. 13 may be used. In general, the pay lines start in the first column and end in the last column, and include one cell per column. However, one or more pay lines could include one or more cells in the same column and may include a vertical pay line.

(45) In some embodiments, upon detecting the occurrence of the trigger condition occurring with the primary game, the control unit **20** executes a bonus feature event algorithm to display a bonus feature event. For example, upon detecting the occurrence of the trigger condition, the control unit **20** may display additional game symbols and/or game animations on the game screen and/or transitions the game screen from the primary game to the bonus feature event.

(46) Referring to FIGS. 5-8, in the illustrated embodiment, the memory **42** stores a game application program **200** that includes computer executable instructions that, when executed by the processor **38**, cause the processor **38** to generate and display the game on the display unit **14** of the gaming machine **10**. In one embodiment, the game application program **200** includes program code **202** and program object data **204** that includes computer executable instructions for implementing a game using the algorithms shown in FIG. 14.

(47) In the illustrated embodiment, the memory **42** stores the game application program **200** and a system application program **206** that includes computer executable instructions that, when executed by the processor **38**, cause the processor **38** to generate and display the game on the display unit **14** of the gaming machine **10**. The game application program **200** provides game specific/front-end functions and the system application program **206** program provides generic/back-end functions, when executed by the processor **38**. In the illustrated embodiment, the game application program **200** and the system application program **206** are implemented on the same operating system. However, it should be noted that these programs may be implemented on different operating system and/or by different processors. In one embodiment, the game application program **200** includes a plurality of software modules including a bet/denomination/payline button listener module **208**, a start button listener module **210**, a credit balance manager module **212**, a sampling manager **214**, a random number generator **216**, a comparison manager **218**, a game result generator **220**, a win evaluator **222**, a game presenter **224**, a game graphics presenter **226**, a game sound presenter **228**, a win indicator **230**, an award provider **232**, an application manager **234**, and an external communicator **236**. The game application program **200** may also include the reel strip data files, the symbol image data files, the variable symbol reel strip data files, the reel stop position data file, the credit prize symbol weight tables, the pay line set, and the payable data file.

(48) The bet/denomination/payline button listener module **208** is a software module for receiving a signal from the bet button, the denomination button, or the payline button which is generated by the button when a player operates the button to select number of bet, denomination, or number of paylines. In response to receiving the signal, the bet/denomination/payline button listener module **208** communicates the occurrence of the signal to application manager **234** for changing bet, denomination, or payline configuration of the game.

(49) The start button listener module **210** is a software module for receiving a signal from the start button which is generated by the button when a player operates the button to start a game. In response to receiving the signal, the start button listener module **210** communicates the occurrence of the signal to application manager **234** for starting the game.

(50) In response to receiving the signal from start button listener module **210**, the application manager **234** requests the sampling manager **214** to obtain necessary number of random numbers from the random number generator **216**.

- (51) The random number generator **216** generates random numbers based on predetermined algorithm of computational random generation method. The random number generator **216** may be a pseudorandom generator. In response to a request from sampling manager **214**, the random number generator **216** returns random number. In some implementations, the random number generator **216** may be implemented in a central server. The random number generator **216** may be implemented as an integrated circuit or hard wired logic.
- (52) The comparison manager **218** compares the current state of the game or each random number with the reel strip data files, the symbol selection data files, the reel stop position data file, the pay line set, and/or the payable data file and specifies corresponding reel layout, stop position, prize symbol, or trigger symbol based on each random number.
- (53) The game result generator **220** generates game result based on selected reel layout, stop positions of each reel, and bonus features.
- (54) The win evaluator **222** evaluates the game result with reference to the pay table.
- (55) The game presenter **224** provides game presentation process with visual and sound so as to form the predetermined game result finally.
- (56) The game graphics presenter **226** provides visual game presentation process on the display so as to form the predetermined game result finally.
- (57) The game sound presenter **228** provides sound presentation process by using sound controller and speakers.
- (58) The win indicator **230** indicates win combinations and payment condition of prize symbol formed in the game result.
- (59) The award provider **232** provides award credit to win meter based on the win evaluation.
- (60) The application manager **234** administrates activity and status of each software module. In addition, the application manager **234** administrates configuration, progress and states of the game application program **200**.
- (61) The external communicator **236** communicates instruction and data with the system application program **206**.
- (62) The credit balance manager module **212** executes a process for decrementing credit balance and incrementing credit balance based on win amount displayed in win meter.
- (63) In the illustrated embodiment, the system application program **206** provides back ground processing and functions other than game specific functions. The system application program **206** includes a plurality of software modules including a system manager **238**, a security manager **240**, a slot management module **242**, a denomination manager **244**, a data logger **246**, a communications manager **248**, a bill acceptor manager **250**, a metering module **252**, and a cashout manager **254**.
- (64) The system application program **206** may also include a game recall file **256**, accounting logs **258**, and meters **260**.
- (65) The system manager **238** is a software module for administrating all of the back ground processing and functions other than game specific functions conducted by the system application program **206**.
- (66) The security manager **240** is a software module for administrating game verification, door security and monitoring security sensors.
- (67) The slot management module **242** is a software module for administrating data accumulation and communicating with external slot information system **62**.
- (68) The denomination manager **244** is a software module for establishing denomination setting of the gaming machine **10**. The denomination setting may include 1 cent, 2 cent, 5 cent, 25 cent, 1 dollar, 5 dollar and the like.
- (69) The data logger **246** is a software module for logging result of each primary game and the free game bonus to the game recall. In addition, the data logger **246** stores error events, bill log, cashout log, ticket log etc. to the accounting log.
- (70) The game recall file **256** is an accumulated data including results of each primary game and

free game bonus. The game recall file **256** is stored in a non-volatile memory.

(71) The accounting logs **258** is an accumulated data including error events, bill log, cashout log, ticket log etc. The accounting logs **258** are stored in a non-volatile memory.

(72) The communications manager **248** is a software module for administrating communication between game application program **200** and system application program **206**. The communications manager **248** also administrates network communication between system application program **206** and external network such as slot management system network, G2S network, gaming server for server based gaming network or VLT system network.

(73) The bill acceptor manager **250** is a software module for administrating the bill acceptor and receives bill information inserted in the bill acceptor. In response to receiving the information from the bill acceptor, the bill acceptor manager **250** communicates with the metering for incrementing credit balance based on the inserted bill.

(74) The metering module **252** is a software module for adjusting values of the meters **260** in response to communication with the game application program **200** via communications manager **248**, the bill acceptor manager **250** or the cashout manager **254**. The meters **260** includes a credit meter for indicating current credit balance on the gaming machine and a win meter for indicating win amount of current game session. The meters further include back ground meters such as coin-in, coin-out, total drop, attendant paid jackpots and/or bill-in. These meters might be implemented as data on the non-volatile memory or hardware meters.

(75) The cashout manager **254** is a software module for administrating cashout procedure. In response to a player's operation on the cashout button, the cashout manager **254** is activated and the gaming machine pay total amount of the credit meter.

(76) Referring to FIGS. **9** and **10**, in one embodiment, the present invention includes a networked server computer system **300** that is configured to deliver the game to one or more client computing devices **302** over the Internet. In the illustrated embodiment, the networked computer system **300** includes an iGaming server system **304** that is coupled in communication with one or more client computing devices **302** via a communications network **306**. The communications network **306** may be any suitable connection, including the Internet, an Intranet, LAN, a virtual private network (VPN), cellular networks, etc. . . . , and may utilize any suitable or combination of technologies including, but not limited to wired and wireless connections, always on connections, connections made periodically, and connections made as needed.

(77) The client computing device **302** may include any suitable device that enables a user to access and communicate with the server system **300** including sending and/or receiving information to and from the server system **300** and displaying information received from the server system **300** to a user. In the illustrated embodiment, the client computing device **302** includes a processor coupled to a memory device. The memory device stores various programs and data that are executed by the processor for operating the client computing device **302**. The client computing device **302** also includes an input device configured to receive operational inputs from the user, and a display device configured to display a graphical user interface. The input device and display device enable a user to interact with the server system **300** via the client computing device **302**. For example, in one embodiment, the client computing device **302** may include, but is not limited to, a desktop computer, a laptop or notebook computer, a tablet computer, smartphone/tablet computer hybrid, a personal data assistant, a handheld mobile device including a cellular telephone, and the like. In one embodiment, the processor of the client computing device **302** may be programmed to function as the control unit **22** of the gaming machine **10**.

(78) In the illustrated embodiment, the client computing device may include a web browser program stored in the memory device. The processor executes the web browser program to display web pages on the display device that includes information received from the server system **300** to enable a user to interact with and operate the server system **300**.

(79) In one embodiment, the client computing device **302** includes a mobile computing device **308**

(shown in FIG. 10) such as, for example, a tablet computer, a smartphone/tablet computer hybrid, a smartphone such as an iPhone™, and the like. The mobile computing device **308** includes a processor coupled to a memory device for storing various programs and data for use in operating the mobile computing device **308**. The mobile computing device **308** may also include a display unit **14** including a touchscreen, one or more video image cameras, one or more speakers, a microphone, at least one input button, and one or more sensors including, but not limited to, a touch ID fingerprint sensor coupled to an input button, a barometer, a three-axis gyro, an accelerometer, proximity sensor, and an ambient light sensor. In addition, the mobile computing device **308** may also include a Wi-Fi antenna, a cellular network antenna, a Bluetooth™ communications device, assisted GPS and GLONASS, a digital compass, and an iBeacon™ microlocation device.

(80) The mobile computing device **308** may be programmed to store and execute mobile computer program applications that display graphical user interfaces on the touchscreen display unit **14** including display area **64** that allows the user to access the server system **300** to retrieve and store information within the server system **300** as well as interact with and operate the server system **300**. In addition, in one embodiment, the server system **300** may install one or more mobile computer application programs in the memory device of the mobile computing device **308**. When initiated by the processor of the mobile computing device **308**, the mobile computer application program causes the processor of the mobile computing device **308** to perform some or all of the functions of the gaming machine **10**.

(81) In the illustrated embodiment, the server system **300** includes one or more remote gaming servers **310**, one or more back-end servers **312**, one or more real money gaming website hosting servers **314**, and one or more social gaming website hosting servers **316**. In the illustrated embodiment, the social gaming website hosting server **316** and the real money gaming website hosting server **314** are programmed to host a website that is accessible by a user via one or more client computing devices **302**. The website hosting servers **314** and **316** execute a website application program that retrieves application code from the back-end server **312** and executes the application code to render one or more webpages on a display device of a client computing device **302** in response to requests received from the user via the client computing device **302** to allow users to interact with the website. The website hosting servers **314** and **316** are configured to generate and display webpages displaying a game. For example, the real money gaming website hosting server **314** is configured to host a real money wagering web site that enables players to convert monetary funds to gaming credits that may be used to place wagers on the game. The social gaming website hosting server **316** is configured to host a social media and/or social gaming website that allows players to receive gaming credits for activities such as purchasing goods and/or services through an e-commerce website, and/or purchase gaming credits that may be used to play the game.

(82) Each back-end server **312** is configured to perform operations to support the functions of the webpages and/or website being displayed by the website hosting servers **314** and **316**. For example, in one embodiment, the back-end servers **312** may include a player account system server that is configured to generate player accounts that include data associated with a player including, but not limited to, player identification information, player financial account information, player gaming credit account information, and/or any suitable player information, that may be used to establish credit meters and allow players to place wagers on the game.

(83) Each remote gaming server **310** includes one or more copies of the game application program **200** stored in a memory device of the remote gaming server **310**. A processor of the remote gaming server **310** is programmed to retrieve and transmit the game application program **200** to one or more back-end servers **312** for use in displaying the game to the user via a webpage being displayed by the web browser program.

(84) In one embodiment, the game application program **200** may include instructions for rendering

the game and executing the game on the client computing device **302**. For example, the game application program **200** may include instructions for generating rendered code, such as, for example HTML code, which may be used by the web browser program of the client computing device **302** for displaying the game. For example, the game application program **200** may include program software code including, but not limited to, HTML, JavaScript, cascade style sheets (CSS), and any suitable programming code that may be used for rendering and operating the game via a web site and/or mobile computer application.

(85) In one embodiment, upon receiving a request from the website hosting servers **314, 316** via the back-end server **312**, the remote gaming server **310** may execute the game application program **200** to operate the game, and execute a render-to-string operation to generate rendered code indicative of the game, such as, for example HTML code, and transmit the rendered code to the back-end server **312**. The back-end server **312** may then transmit the rendered code to the corresponding website hosting servers **314, 316** for use in displaying the game on the website. As the player plays the game, the remote gaming server **310** may execute the game application program **200** for each instance of the game, and transmit rendered code to the back-end servers **312**.

(86) In another embodiment, the remote gaming server **310** may transmit the game application program **200** to the back-end server **312** and/or the website hosting servers **314, 316**. The back-end server **312** and/or the website hosting servers **314, 316** may then execute the game application program **200** to initiate the instances of the game and execute render-to-string operations to generate rendered code indicative of the game.

(87) In yet another embodiment, the back-end server **312** may receive a request to initiate the game from a mobile computing device **308** executing the mobile computer application program. Upon receiving the request, the back-end server **312** may access the game application program **200** and execute a render-to-string operation to generate rendered code indicative of the game and transmit the rendered code to the mobile computing device **308**. In one embodiment, the back-end server **312** may continuously execute the game application program **200** to generate each instance of the game using a random number generator of the back-end server **312** based on input received from the mobile computing device **308** and generate and transmit rendered code for each instance of the game to the mobile computing device **308**. In another embodiment, the back-end server **312** may execute a partial-render operation and generate partially-rendered code of the game using the game application program **200**, and transmit the partially rendered code of the game and object data of game assets to the mobile computing device **308**. The partially rendered code includes instructions for generating rendered code using the game assets and a random number generator of the mobile computing device for generating and displaying the game on the mobile computing device **308** using the mobile computer application program.

(88) In one embodiment, the game application program **200** may be stored on several different servers. The game code on these servers is used to distribute game content to social or real money gaming websites and mobile applications. The distribution method is very flexible. For example, the game code and/or game application program **200** including game code and game object assets may be stored on a remote gaming server **310**. One remote gaming server **310** may be connected to one or more back-end server **312**.

(89) Each back-end server **312** is configured to distribute the games to one or more websites or mobile applications. Players connect to these websites/mobile applications with the client devices or mobile devices and have access to the game content. A copy of game application program **200** including game code and game object assets is stored on the remote gaming server **310** for each back-end server **312** that is connected to the remote gaming server **310** and that distributes the game. For example, if one remote gaming server **310** is connected to two back-end servers **312**, which is connected to three website hosting servers **314, 316** that distribute the game, the remote gaming server **310** would store two copies of the game application program **200** including game code and game object assets for the game (e.g., one copy for each back-end server **312**).

(90) For example, the server system **300** may be configured to implement the game on a mobile application such as, for example, “my KONAMI Slots™” mobile application available in Apple iOS™, Google Android™, and Amazon Kindle™ operating platforms, or on social-media websites such as the “my KONAMI Slots™” available on Facebook™. In one embodiment, the mobile application may download the game code from remote gaming server **310** via the real money gaming site **314** or the social gaming site **316** and execute the game code on the client computing device **302**. In this embodiment, the game code may provide game specific/front-end function when executed by the processor of the client computing device, and the back end system **312** may provide generic/back-end function.

(91) FIGS. **14-16** are flow charts of methods **400**, **500**, and **600** illustrating the algorithms included in the game application program **200** and performed by the processor **38** when executing the game application program **200** for operating the gaming machine **10** and/or iGaming server system **300** to generate and render animated sequences of computer-generated images on graphic user interface screens to display the game to the player. The methods include a plurality of steps. Each method step may be performed independently of, or in combination with, other method steps. Portions of the methods may be performed by any one of, or any combination of, the components of the gaming machine **10** and/or iGaming server system **300**.

(92) For example, the game application program **200** may include computer-executable instructions including algorithms for providing the Rising Diamond™ Pink Edition™ and the Rising Diamond™ Blue Edition™ slot games offered by Konami Gaming™ Inc. which include the Rising Reels™ feature event and the Rising Diamonds™ bonus feature event.

(93) In the illustrated embodiment, the game application program **200** includes computer instructions for generating the primary game **68** and the bonus feature event **69**. In general, the bonus feature event **69** is executed when a triggering event is detected during the primary game **68**. In one embodiment, the primary game **68** includes a reel-type game that includes a plurality of virtual reels that spin and stop to display the outcomes of the primary game. In other embodiments, the primary game **68** may include a playing card game, a bingo game, a Keno game, and/or any suitable casino type wagering game.

(94) In the illustrated embodiment, the control unit **20** is programmed to execute the algorithm illustrated in methods **400** for executing the game and rendering animated sequences of computer-generated images shown in FIGS. **17-27** displaying the game on the display unit **14**. In method step **402**, the processor **38** of the control unit **20** generates the virtual reels **70-78** for use with an instance of the primary game **68**, and displays the virtual reels **70-78** in the grid **80**.

(95) In method step **404**, the processor **38** receives a signal to initiate an instance of the game **66** and generates the virtual reels **70-78** for use during the instance of the game **66**. For example, the processor **38** may receive a signal from the display unit **14** and/or the operation unit **32** indicating a player's request to initiate an instance or play of the game **66**. In one embodiment, the player may transmit a request to initiate an instance or play of game **66** by depressing a corresponding “spin” button on the operation unit **32** by touching a corresponding button icon being displayed on a touchscreen of the display unit **14**. Upon receiving a signal from the display unit **14** and/or the operation unit **32**, the processor **38** initiates the instance of the game **66** by generating the virtual reels **70-78** for use in the primary game **68**.

(96) In method steps **406-416**, the processor **38** randomly determines an outcome of an instance of the primary game **68** and spins the virtual reels **70** to **78** and sequentially stops the virtual reels **70** to **78** to display the randomly generated outcome including a game symbol being displayed in each cell **82** of the grid **80**. For example, in one embodiment, the processor **38** starts spinning each virtual reel **70** to **78**, obtains random numbers from the random number generator, and determines a stop position of each virtual reel **70** to **78** based on the random numbers and the reel stop position data file **108**. In the illustrated embodiment the reel stop position data file **108** includes a range of random numbers associated with each symbol position in the sequence of symbol positions

associated with the virtual reel. In one embodiment, the processor may obtain a random number for each simulate virtual reel **70** to **78**, i.e., five random numbers. The processor **38** then establishes a reel stop counter, “i”, and sets the reel stop counter, i, equal to 1. The processor **38** then identifies the ith virtual reel strip associated with the stop counter, i, and stops the identified virtual reel strip to display the corresponding symbols in the corresponding cells **82** associated with the identified virtual reel strip. The processor **38** then increments the reel stop counter, i, by 1, i.e., $i=i+1$, and repeats the process of identifying the virtual reel strip associated with the incremented reel stop counter and stopping the identified virtual reel. This process continues until each virtual reel has been stopped. In this embodiment, for example, the virtual reels are numbered 1-5. In one embodiment, during the reel spin, the player may initiate the stopping of the reels by depressing the spin button, which enables the player to accelerate game play.

(97) In method step **418**, the processor **38** determines whether a trigger condition has been detected in the outcome of the primary game **68**. If a trigger condition is detected during the primary game **68**, the processor **38** initiates the bonus feature event in method step **420**.

(98) In method step **422**, upon completion of the primary game **68** and/or the bonus feature event, the processor **38** determines and provides any award associated with the outcome of the primary game **68** and/or the bonus feature event. For example, the processor **38** may determine whether any winning combination of symbols is displayed in the outcome of the instance of the primary game **68** and/or the bonus feature event provide an award by increasing a balance of the credit meter based on the amount of the award.

(99) In some embodiments, the control unit **20** is programmed to execute the algorithm illustrated in methods **500** and **600** for executing the game and rendering animated sequences of computer-generated images shown in FIGS. **17-27** displaying a computer-animated primary reel game **68** (shown in FIGS. **3**, and **17-21**) and a computer-animated bonus game **69** (shown in FIGS. **22-23**).

(100) For example, in method step **502**, the processor **38** renders a primary game screen **120** (shown in FIGS. **17-21**) on the display device **14** including a plurality of virtual reels **70**, **72**, **74**, **76**, **78** displayed within a reel window **122**. The reel window **122** displays a grid **80** including a plurality of cells **82** arranged in a plurality of rows and columns. Each virtual reel **70**, **72**, **74**, **76**, **78** is displayed in a corresponding column.

(101) In method step **504**, the processor **38** initiates an instance of a computer-animated primary reel game **68** by animating the plurality of virtual reels **70**, **72**, **74**, **76**, **78** to simulate spinning the plurality of virtual reels through the reel window **122** and sequentially stopping the plurality of virtual reels **70**, **72**, **74**, **76**, **78** to display a game outcome (shown in FIG. **19**).

(102) In method step **506**, the processor **38** determines whether a special symbol **124** appears in the game outcome within the reel window **122**. Upon determining the game outcome displays a special symbol **124** within the reel window, the processor **38** executes method step **508** to reset a spin counter, which counts the number of consecutive game outcomes that have not included a special symbol appearing the reel window **122** with the reels stopped. For example, because the current game outcome displays a special symbol **124**, the spin counter is reset to zero. The processor **38** then executes method step **510** and determines whether a trigger condition has occurred during the computer-animated primary reel game **68**. If a trigger condition has occurred, the processor **38** executes methods step **512** and initiates the computer-animated bonus game **69** using method **600**. If a trigger condition has not occurred, the processor **38** executes methods step **514**.

(103) In method step **514**, upon determining the game outcome displays a special symbol **124** within the reel window **122**, the processor **38** transforms the reel window **122** by animating the reel window **122** to increase a size of the reel window **122** to display an additional row of cells **82** within the reel window **122** for use in a subsequent instance of the computer-animated primary reel game **68**, as shown in FIGS. **20**, **21**, and **24**. In the illustrated embodiment, the processor **38** animates the reel window **122** to increase a size of the reel window **122** as the plurality of virtual reels **70**, **72**, **74**, **76**, **78** are animated to simulate spinning. In other embodiments, the processor **38**

animates the reel window **122** to increase the size of the reel window **122** before the virtual reels **70, 72, 74, 76, 78** are animated to simulate spinning. For example, as shown in FIG. **20**, the processor **38** may display the reel window with an initial cell arrangement including a 4×5 cell arrangement including a 4×4×4×4×4 cell grid. Upon determining the game outcome displays a special symbol **124** within the reel window **122**, the processor **38** animates the reel window **122** to increase a size of the reel window **122** from a 4×5 cell arrangement to a 5×5 cell arrangement including a 5×5×5×5×5 cell grid, thereby increasing a number of potential winning outcomes. In some embodiments, the processor **38** may be programmed to animate a winning ways image icon **126** (shown in FIGS. **20** and **21**) across the reel window **122** indicating a number of potential winning combinations associated with the increased size of the reel window **122**. The winning ways image icon **126** indicates a number of potential winning combinations associated with the current cell arrangement, for example, a 3×5 cell arrangement provides a 243 ways game; a 4×5 cell arrangement provides a 1024 ways game; a 5×5 cell arrangement provides a 3125 ways game, a 6×5 cell arrangement provides a 7776 ways game; and a 7×5 cell arrangement provides a 16807 ways game. The processor **38** may temporarily animate the winning ways image icon **126** as the reel window **122** increases in size, and remove the winning ways image icon **126** from the primary game screen **120** once the reel window **122** has reached the increased height to enable the spinning virtual reels **70, 72, 74, 76, 78** to be viewed through the increased reel window **122**.

(104) In some embodiments, the processor **38** may be programmed to determine a number of special symbols **124** appearing in the game outcome and transform the reel window **122** by animating the reel window **122** to increase the size of the reel window **122** to display an additional number of rows of cells equal to the number of special symbols **124** appearing in the game outcome. For example, as shown in FIG. **24** the processor **38** may transform the reel window **122** by animating the reel window **122** to increase the size of the reel window **122** to a 4×5, 5×5, 6×5, or 7×5 cell arrangement based on the number of special symbols **124** appearing in the game outcome. As shown in FIG. **20**, the processor **38** may increase the reel window **122** by one row upon detecting one special symbol **124** appearing in the game outcome. For example, if the current reel window includes a 3×5 cell arrangement, the processor **38** will increase the reel window **122** to a 4×5 cell arrangement; if the current reel window **122** includes a 4×5 cell arrangement, the processor **38** will increase the reel window **122** to a 5×5 cell arrangement; if the current reel window **122** includes a 5×5 cell arrangement, the processor **38** will increase the reel window **122** to a 6×5 cell arrangement (i.e., a 6×6×6×6×6 cell grid); and if the current reel window **122** includes a 6×5 cell arrangement, the processor **38** will increase the reel window **122** to a 7×5 cell arrangement (i.e., a 7×7×7×7×7 cell grid).

(105) The processor **38** may also be programmed to increase the reel window **122** by two rows upon detecting two special symbols **124** appearing in the game outcome. For example, if the current reel window includes a 3×5 cell arrangement, the processor **38** will increase the reel window **122** to a 5×5 cell arrangement; if the current reel window **122** includes a 4×5 cell arrangement, the processor **38** will increase the reel window **122** to a 6×5 cell arrangement; and if the current reel window **122** includes a 5×5 cell arrangement, the processor **38** will increase the reel window **122** to a 7×5 cell arrangement.

(106) The processor **38** may also be programmed to increase the reel window **122** by three rows upon detecting three special symbols **124** appearing in the game outcome. For example, if the current reel window includes a 3×5 cell arrangement, the processor **38** will increase the reel window **122** to a 6×5 cell arrangement; and if the current reel window **122** includes a 4×5 cell arrangement, the processor **38** will increase the reel window **122** to a 7×5 cell arrangement.

(107) Upon animating the reel window **122** to increase the size of the reel window **122** as the plurality of virtual reels **70, 72, 74, 76, 78** are animated to simulate spinning, the processor **38** then returns to method step **504** and animates the plurality of virtual reels **70, 72, 74, 76, 78** to sequentially stop the plurality of virtual reels **70, 72, 74, 76, 78** to display a subsequent game

outcome.

(108) In method step **506**, upon determining the game outcome does not display a special symbol **124** within the reel window **122**, the processor **38** executes method step **516** and increments the spin counter by one, to count the number of consecutive game outcomes that have not included a special symbol **124** appearing the reel window **122** with the reels stopped. In method step **518**, the processor **38** determines a number of consecutive previous game outcomes that did not include the special symbol **124** appearing the reel window **122** with the reels stopped. If the processor **38** determines the number of consecutive previous game outcomes that did not include the special symbol **124** is equal to a predefined number of game outcomes, the processor **38** executes method step **520** and transforming the reel window by animating the reel window **122** to decrease the size of the reel window **122** (shown in FIG. **20**) to display a fewer number of rows of cells within the reel window **122** for use in the subsequent instance of the computer-animated primary reel game.

(109) If the processor **38** determines the number of consecutive previous game outcomes that did not include the special symbol is not equal to a predefined number of game outcomes, the processor **38** returns to method step **504** and animates the plurality of virtual reels **70, 72, 74, 76, 78** to sequentially stop the plurality of virtual reels **70, 72, 74, 76, 78** to display a subsequent game outcome without adjusting a size of the reel window **122**. For example, in some embodiments, the processor **38** renders a spin counter image **128** (shown in FIG. **18**) at a lower portion of the reel window **122** to notify the player of a number of reel spins remaining before the reel window **122** is reduced in size and indicating a number of potential winning combinations (e.g. 243 ways, 1024 ways, 3125 ways, 7776 ways, or 16807 ways) associated with the current size of the reel window **122**

(110) For example, at method step **518**, the processor **38** may determine whether the spin counter is equal to three, which indicates three consecutive instances of the primary game **68** in which the game outcome did not include the special symbol **124** appearing the reel window **122** with the reels stopped. Upon determining that three consecutive game outcomes did not display the special symbol **124**, the processor **38** transforms the reel window **122** by animating the reel window **122** to decrease a size of the reel window **122** to removes a row of cells being displayed within the reel window **122** for use in a subsequent instance of the computer-animated primary reel game **68**, as shown in FIGS. **20** and **24**. In the illustrated embodiment, the processor **38** animates the reel window **122** to decrease a size of the reel window **122** as the plurality of virtual reels **70, 72, 74, 76, 78** are animated to simulate spinning. In other embodiments, the processor **38** may decrease the size of the reel window **122** before the virtual reels **70, 72, 74, 76, 78** are animated to simulate spinning.

(111) For example, if the current reel window **122** includes a 7×5 cell arrangement, the processor **38** will decrease the reel window **122** to a 6×5 cell arrangement; if the current reel window **122** includes a 6×5 cell arrangement, the processor **38** will decrease the reel window **122** to a 5×5 cell arrangement; if the current reel window **122** includes a 5×5 cell arrangement, the processor **38** will decrease the reel window **122** to a 4×5 cell arrangement; and if the current reel window **122** includes a 4×5 cell arrangement, the processor **38** will decrease the reel window **122** to a 3×5 cell arrangement. If the current reel window **122** includes a 3×5 cell arrangement, the processor **38** will not decrease the reel window **122**, and will initiate the subsequent instance of the computer-animated primary reel game **68** with the 3×5 cell arrangement.

(112) With this function, symbol array is extended by a row when a Wild symbol occurs in a previous game. And the symbol array is reduced by a row when no wild occurs for three games. Furthermore, free games are awarded when a Wild symbol occurs in the symbol array which already reached maximum size. For example, when a wild occurs, the reel widow increases for the next game. When a wild doesn't appear **3** games in a row, the reel widow will be decreased for the next game. If reel widow size reaches to the top, Rising Diamond Bonus will be awarded.

(113) In method step **510**, if the processor **38** detects a trigger condition occurring during the

computer-animated primary reel game **68**, the processor **38** executes methods step **512** and initiates the computer-animated bonus game **69** using method **600**. For example, the processor **38** may detect the trigger condition when: three or more special symbols **124** appear in the reel window **122** having a 5×5 cell arrangement; two or more special symbols **124** appear in the reel window **122** having a 6×5 cell arrangement; or one or more special symbols **124** appear in the reel window **122** having a 7×5 cell arrangement. In one embodiment, as shown in FIG. **21**, upon detecting the trigger condition during the computer-animated primary reel game **68**, the processor **38** may animate the reel window **122** to increase the size of the reel window **122** until the maximum size of the reel window **122** is reached, i.e., a 7×5 cell arrangement, temporarily displaying each corresponding winning ways image icon **126** associated with each cell arrangement.

(114) In the illustrated embodiment, the processor **38** executes method **600** to initiate the computer-animated bonus game **69** (shown in FIGS. **22** and **23**) upon detecting a trigger condition during the computer-animated primary reel game **68**. In method step **602**, the processor **38** renders a bonus selection game screen **130** on the display device **14** including a preview reel window **132** displayed within an upper portion of the bonus selection game screen **130** and a plurality of player selectable feature icons **134** displayed within a lower portion of the bonus selection game screen **130**. For example, when initiating the computer-animated bonus game **69**, the processor **38** displays an intermediate animation sequence to replace the primary game screen **120** with the bonus selection game screen **130** on the display device **14**.

(115) The preview reel window **132** is displayed including a plurality of active cells **136** and a plurality of inactive cells **138**. For example, the preview reel window **132** is initially displayed with a plurality of active cells **136** arranged in a 7×5 cell arrangement, and three additional rows of inactive cells **138** displayed above the active cells **136**. Each selected player selectable feature icon **134** is associated with a bonus feature. For example, in the illustrated embodiment, the computer-animated bonus game **69** includes a plurality of bonus features including a reel window increase bonus feature **140** that transforms a row of inactive cells **138** to active cells **136** displayed in the preview reel window **132**; a free game feature **142** that increases a number of free games available during the computer-animated bonus game **69**; a multiplier feature **144** that increase a multiplier value applied to winning combinations during the computer-animated bonus game **69**; and an extra-pick feature **146** that increases a number of player selections available during the computer-animated bonus game **69**. For example, the computer-animated bonus game **69** may initially provide two picks, i.e., player selections, to the player. If the extra-pick feature **146** is selected during the bonus game **69**, the processor **38** adds a number of picks to a remaining number of pick available to the player.

(116) In method step **604**, the processor **38** receives a player's selection of a player selectable feature icon **134** and animates the selected player selectable feature icon **134** to reveal an associated bonus feature. For example, the processor **38** may receive a player's selection of a player selectable feature icon **134** via the touchscreen of the display device **14**. Upon receiving the player's selection, the processor **38** animates the selected player selectable feature icon **134** to reveal the associated bonus feature.

(117) In method steps **606-620**, the processor **38** determines the bonus feature associated with the selected player selectable feature icon **134** and performs the corresponding animated sequence. For example, if the processor **38** determines the revealed bonus feature is a reel window increase bonus feature **140**, the processor **38** executes method step **620** and transforms a row of inactive cells **138** to active cells **136** to increase the number of active cells **136** displayed within the preview reel window **132** upon determining the revealed bonus feature is a reel window increase bonus feature **140**. The processor **38** may also animate a reel arrangement image icon **148** across the preview reel window **132** indicating a reel arrangement associated with the number of active cells **136**.

(118) If the processor **38** determines the revealed bonus feature is a multiplier feature **144** that increase a multiplier value applied to winning combinations during the computer-animated bonus

game **69**, the processor **38** executes method step **610** and animates a multiple image icon **150** to increase a multiplier value indicated on the multiple image icon **150**.

(119) If the processor **38** determines the revealed bonus feature is free game feature **142** that increases a number of free games available during the computer-animated bonus game **69**, the processor **38** executes method step **614** and animates a free games image icon **152** to indicate the number of free games available in the bonus game **69**.

(120) If the processor **38** determines the revealed bonus feature is an extra-pick feature **146** that increases a number of player selections available during the computer-animated bonus game **69**, the processor **38** executes method step **618** and animates a picks remaining image icon **154** to indicate an increase in the number of remaining player selections in the bonus game **69**.

(121) In method step **622**, the processor **38** determines if any player selections remain in the bonus game **69**.

(122) If there are remaining player selections available, the processor **38** returns to method step **604** to receive another player selection of a player selectable feature icon **134**. If there are no remaining player selections available, the processor **38** executes method step **624** and transforms the bonus selection game screen **130** to display a bonus feature game screen **156** including a plurality of bonus reels **158** being displayed within a bonus reel window **160** that includes the reel arrangement previously displayed within the preview reel window **132**. For example, as shown in FIG. **23**, if the preview reel window **132** displays a 7×5 cell arrangement when the processor **38** determines there are no remaining player selections available in the pick game, the processor **38** then animates the bonus feature game screen **156** on the display device **14** including a bonus reel window **160** having a 7×5 cell arrangement. The processor **38** then initiates a free game instance of the bonus game by animating the plurality of bonus reels **158** to simulate spinning the plurality of bonus reels through the bonus reel window **160** and sequentially stopping the plurality of bonus reels to display a bonus game outcome.

(123) Exemplary embodiments of a gaming device, a gaming system, and a method of providing an award to a player are described above in detail. The gaming device, system, and method are not limited to the specific embodiments described herein, but rather, components of the gaming device and/or system and/or steps of the method may be utilized independently and separately from other components and/or steps described herein. For example, the gaming device may also be used in combination with other gaming systems and methods, and is not limited to practice with only the gaming device as described herein. Rather, an exemplary embodiment can be implemented and utilized in connection with many other gaming system applications.

(124) A controller, computing device, or computer, such as described herein, includes at least one or more processors or processing units and a system memory. The controller typically also includes at least some form of computer readable media. By way of example and not limitation, computer readable media may include computer storage media and communication media. Computer storage media may include volatile and nonvolatile, removable and non-removable media implemented in any method or technology that enables storage of information, such as computer readable instructions, data structures, program modules, or other data. Communication media typically embody computer readable instructions, data structures, program modules, or other data in a modulated data signal such as a carrier wave or other transport mechanism and include any information delivery media. Those skilled in the art should be familiar with the modulated data signal, which has one or more of its characteristics set or changed in such a manner as to encode information in the signal. Combinations of any of the above are also included within the scope of computer readable media.

(125) The order of execution or performance of the operations in the embodiments of the invention illustrated and described herein is not essential, unless otherwise specified. That is, the operations described herein may be performed in any order, unless otherwise specified, and embodiments of the invention may include additional or fewer operations than those disclosed herein. For example,

it is contemplated that executing or performing a particular operation before, contemporaneously with, or after another operation is within the scope of aspects of the invention.

(126) In some embodiments, a processor, as described herein, includes any programmable system including systems and microcontrollers, reduced instruction set circuits (RISC), application specific integrated circuits (ASIC), programmable logic circuits (PLC), and any other circuit or processor capable of executing the functions described herein. The above examples are exemplary only, and thus are not intended to limit in any way the definition and/or meaning of the term processor.

(127) This written description uses examples to disclose the invention, including the best mode, and also to enable any person skilled in the art to practice the invention, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the invention is defined by the claims, and may include other examples that occur to those skilled in the art. Other aspects and features of the present invention can be obtained from a study of the drawings, the disclosure, and the appended claims. The invention may be practiced otherwise than as specifically described within the scope of the appended claims. It should also be noted, that the steps and/or functions listed within the appended claims, notwithstanding the order of which steps and/or functions are listed therein, are not limited to any specific order of operation.

(128) Although specific features of various embodiments of the invention may be shown in some drawings and not in others, this is for convenience only. In accordance with the principles of the invention, any feature of a drawing may be referenced and/or claimed in combination with any feature of any other drawing.

Claims

1. A gaming machine, comprising: a cabinet; a display device mounted to the cabinet; and a control unit operably coupled to the display device, the control unit including a processor programmed to execute an algorithm to render an animated sequence of computer-generated images on the display device including the steps of: rendering a primary game screen on the display device including a plurality of virtual reels displayed within a reel window including a plurality of cells arranged in a plurality of rows and columns, each virtual reel being displayed in a corresponding column; and initiating an instance of a computer-animated primary reel game by: animating the plurality of virtual reels to simulate spinning the plurality of virtual reels through the reel window and sequentially stopping the plurality of virtual reels to display a game outcome; upon determining the game outcome displays a special symbol within the reel window: transforming the reel window by animating the reel window to increase a size of the reel window to display an additional row of cells within the reel window for use in a subsequent instance of the computer-animated primary reel game; and upon determining the game outcome does not display the special symbol: transforming the reel window by animating the reel window to decrease the size of the reel window to display a fewer number of rows of cells within the reel window for use in the subsequent instance of the computer-animated primary reel game.
2. The gaming machine of claim 1, wherein the processor is programmed to execute the algorithm including the steps of: upon determining the game outcome displays the special symbol: determining a number of special symbols appearing in the game outcome; and transforming the reel window by animating the reel window to increase the size of the reel window to display an additional number of rows of cells equal to the number of special symbols appearing in the game outcome.
3. The gaming machine of claim 2, wherein the processor is programmed to execute the algorithm including the steps of: transforming the reel window by animating the reel window to increase the size of the reel window to at least one of a 4×5, 5×5, 6×5, and 7×5 cell arrangement based on the number of special symbols appearing in the game outcome.

4. The gaming machine of claim 2, wherein the processor is programmed to execute the algorithm including the steps of: animating a winning ways image icon across the reel window indicating a number of potential winning combinations associated with the increased size of the reel window.
5. The gaming machine of claim 1, wherein the processor is programmed to execute the algorithm including the steps of: upon determining the game outcome does not display the special symbol: determining a number of consecutive previous game outcomes that did not include the special symbol; and animating the reel window to decrease the size of the reel window upon determining the number of consecutive previous game outcomes that did not include the special symbol is equal to a predefined number of game outcomes.
6. The gaming machine of claim 1, wherein the processor is programmed to execute the algorithm including the steps of: initiating a computer-animated bonus game upon detecting a trigger condition during the computer-animated primary reel game by: rendering a bonus selection game screen on the display device including: a preview reel window displayed within an upper portion of the bonus selection game screen and including a plurality of active cells and a plurality of inactive cells; and a plurality of player selectable feature icons displayed within a lower portion of the bonus selection game screen.
7. The gaming machine of claim 6, wherein the processor is programmed to execute the algorithm including the steps of: receiving a player's selection of a player selectable feature icon; animating the selected player selectable feature icon to reveal an associated bonus feature; transforming a row of inactive cells to active cells to increase the number of active cells displayed within the preview reel window upon determining the revealed bonus feature is a reel window increase bonus feature; and animating a reel arrangement image icon across the preview reel window indicating a reel arrangement associated with the number of active cells.
8. The gaming machine of claim 7, wherein the processor is programmed to execute the algorithm including the steps of: transforming the bonus selection game screen to display a bonus feature game screen including a plurality of bonus reels being displayed within a bonus reel window including the reel arrangement previously displayed within the preview reel window; and animating the plurality of bonus reels to simulate spinning the plurality of bonus reels through the bonus reel window and sequentially stopping the plurality of bonus reels to display a bonus game outcome.
9. A method of operating a gaming machine including a cabinet, a display device mounted to the cabinet, and a control unit including a processor operably coupled to the display device, the method including the processor performing an algorithm to render an animated sequence of computer-generated images on the display device including the steps of: rendering a primary game screen on the display device including a plurality of virtual reels displayed within a reel window including a plurality of cells arranged in a plurality of rows and columns, each virtual reel being displayed in a corresponding column; and initiating an instance of a computer-animated primary reel game by: animating the plurality of virtual reels to simulate spinning the plurality of virtual reels through the reel window and sequentially stopping the plurality of virtual reels to display a game outcome; upon determining the game outcome displays a special symbol within the reel window: transforming the reel window by animating the reel window to increase a size of the reel window to display an additional row of cells within the reel window for use in a subsequent instance of the computer-animated primary reel game; and upon determining the game outcome does not display the special symbol: transforming the reel window by animating the reel window to decrease the size of the reel window to display a fewer number of rows of cells within the reel window for use in the subsequent instance of the computer-animated primary reel game.
10. The method of claim 9, including the processor performing the algorithm steps of: upon determining the game outcome displays the special symbol: determining a number of special symbols appearing in the game outcome; and transforming the reel window by animating the reel window to increase the size of the reel window to display an additional number of rows of cells

equal to the number of special symbols appearing in the game outcome.

11. The method of claim 10, including the processor performing the algorithm steps of: transforming the reel window by animating the reel window to increase the size of the reel window to at least one of a 4×5, 5×5, 6×5, and 7×5 cell arrangement based on the number of special symbols appearing in the game outcome.

12. The method of claim 10, including the processor performing the algorithm steps of: animating a winning ways image icon across the reel window indicating a number of potential winning combinations associated with the increased size of the reel window.

13. The method of claim 9, including the processor performing the algorithm steps of: upon determining the game outcome does not display the special symbol: determining a number of consecutive previous game outcomes that did not include the special symbol; and animating the reel window to decrease the size of the reel window upon determining the number of consecutive previous game outcomes that did not include the special symbol is equal to a predefined number of game outcomes.

14. The method of claim 9, including the processor performing the algorithm steps of: initiating a computer-animated bonus game upon detecting a trigger condition during the computer-animated primary reel game by: rendering a bonus selection game screen on the display device including: a preview reel window displayed within an upper portion of the bonus selection game screen and including a plurality of active cells and a plurality of inactive cells; and a plurality of player selectable feature icons displayed within a lower portion of the bonus selection game screen.

15. The method of claim 14, including the processor performing the algorithm steps of: receiving a player's selection of a player selectable feature icon; animating the selected player selectable feature icon to reveal an associated bonus feature; transforming a row of inactive cells to active cells to increase the number of active cells displayed within the preview reel window upon determining the revealed bonus feature is a reel window increase bonus feature; and animating a reel arrangement image icon across the preview reel window indicating a reel arrangement associated with the number of active cells.

16. The method of claim 15, including the processor performing the algorithm steps of: transforming the bonus selection game screen to display a bonus feature game including a plurality of bonus reels being displayed within a bonus reel window including the reel arrangement previously displayed within the preview reel window; and animating the plurality of bonus reels to simulate spinning the plurality of bonus reels through the bonus reel window and sequentially stopping the plurality of bonus reels to display a bonus game outcome.

17. A non-transitory computer-readable storage media having computer-executable instructions embodied thereon to operate a gaming machine including a cabinet, a display device mounted to the cabinet, and a control unit including a processor operably coupled to the display device, when executed by the processor the computer-executable instructions cause the processor to perform an algorithm to render an animated sequence of computer-generated images on the display device including the steps of: rendering a primary game screen on the display device including a plurality of virtual reels displayed within a reel window including a plurality of cells arranged in a plurality of rows and columns, each virtual reel being displayed in a corresponding column; and initiating an instance of a computer-animated primary reel game by: animating the plurality of virtual reels to simulate spinning the plurality of virtual reels through the reel window and sequentially stopping the plurality of virtual reels to display a game outcome; upon determining the game outcome displays a special symbol within the reel window: transforming the reel window by animating the reel window to increase a size of the reel window to display an additional row of cells within the reel window for use in a subsequent instance of the computer-animated primary reel game; and upon determining the game outcome does not display the special symbol: transforming the reel window by animating the reel window to decrease the size of the reel window to display a fewer number of rows of cells within the reel window for use in the subsequent instance of the computer-

animated primary reel game.

18. The non-transitory computer-readable storage media of claim 17, wherein the computer-executable instructions cause the processor to perform the algorithm including the steps of: upon determining the game outcome displays the special symbol; determining a number of special symbols appearing in the game outcome; and transforming the reel window by animating the reel window to increase the size of the reel window to display an additional number of rows of cells equal to the number of special symbols appearing in the game outcome.

19. The non-transitory computer-readable storage media of claim 17, wherein the computer-executable instructions cause the processor to perform the algorithm including the steps of: upon determining the game outcome does not display the special symbol; determining a number of consecutive previous game outcomes that did not include the special symbol; and animating the reel window to decrease the size of the reel window upon determining the number of consecutive previous game outcomes that did not include the special symbol is equal to a predefined number of game outcomes.

20. The non-transitory computer-readable storage media of claim 17, wherein the computer-executable instructions cause the processor to perform the algorithm including the steps of: initiating a computer-animated bonus game upon detecting a trigger condition during the computer-animated primary reel game by: rendering a bonus selection game screen on the display device including: a preview reel window displayed within an upper portion of the bonus selection game screen and including a plurality of active cells and a plurality of inactive cells; and a plurality of player selectable feature icons displayed within a lower portion of the bonus selection game screen; receiving a player's selection of a player selectable feature icon; animating the selected player selectable feature icon to reveal an associated bonus feature; transforming a row of inactive cells to active cells to increase the number of active cells displayed within the preview reel window upon determining the revealed bonus feature is a reel window increase bonus feature; and animating a reel arrangement image icon across the preview reel window indicating a reel arrangement associated with the number of active cells.

21. The non-transitory computer-readable storage media of claim 20, wherein the computer-executable instructions cause the processor to perform the algorithm including the steps of: transforming the bonus selection game screen to display a bonus feature game including a plurality of bonus reels being displayed within a bonus reel window including the reel arrangement previously displayed within the preview reel window; and animating the plurality of bonus reels to simulate spinning the plurality of bonus reels through the bonus reel window and sequentially stopping the plurality of bonus reels to display a bonus game outcome.
